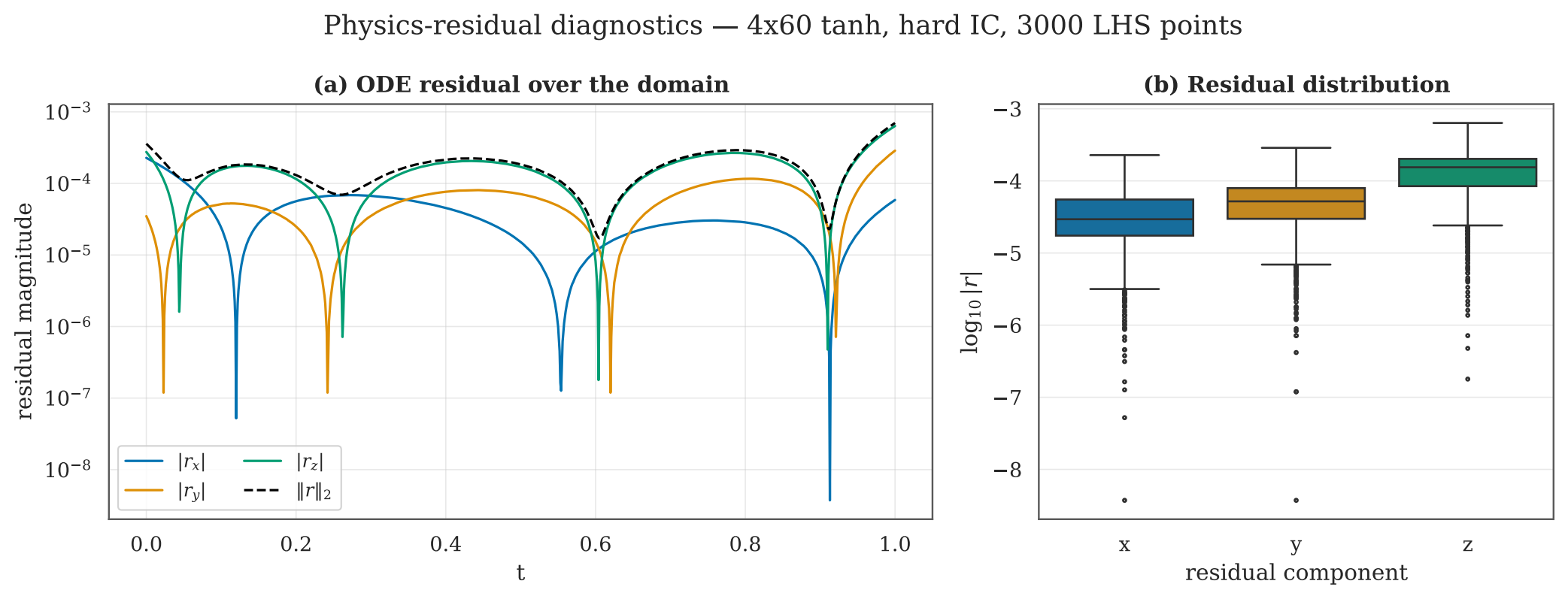
Physics-residual diagnostics — 4x60 tanh, hard IC, 3000 LHS points
(a) ODE residual over the domain
residual magnitude
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
$|r_x|$
$|r_y|$
$|r_z|$
$\|r\|_2$
0.0
0.2
0.4
0.6
0.8
1.0
t
(b) Residual distribution
$\log_{10}|r|$
−3
−4
−5
−6
−7
−8
x
y
z
residual component